Reference Products for Environment and Trade

Previous Proficiency Tests

The IAEA Terrestrial Environmental Radiochemistry Laboratory organizes annual proficiency test (PT) and interlaboratory comparison (ILC) exercises for Member State and ALMERA laboratories. These exercises are designed to monitor and demonstrate their performance and analytical capabilities, and to identify gaps and problem areas where further development is needed.

The annual exercises contain varied sets of environmental samples to be characterized by participating laboratories for anthropogenic and natural alpha, beta and gamma emitting radionuclides. Participants can report their measurement results on radionuclides of their choice; it is not compulsory to report measurement results on all requested radionuclides and/or samples. Individual reports containing the evaluation on the measurement results are available for participants to download within a few months after the reporting deadline.

Current Proficiency Test Exercise

The IAEA-TERC-2025 Worldwide Proficiency Test Exercise application has closed.

Should you have any questions or need further information, please do not hesitate to contact the Proficiency Test Team at Proficiency-Tests.Contact-Point@iaea.org.

Previous Proficiency Tests

IAEA-TERC-2024-01 Worldwide Proficiency Test

Worldwide Proficiency Test on the determination of anthropogenic and natural radionuclies in water, bauxite, sediment, and simulated contaminated surface samples.

- Sample 1: Spiked Water, Anthropogenic and natural radionuclides, gross alpha and beta (GAB), uranium (trace level)
- Sample 3: Sediment, naturally occurring gamma emitting radionuclides

- Sample 4: Bauxite, naturally occurring gamma emitting radionuclides
- Sample 5: Simulated contaminated surface sample, gamma and beta emitting radionuclides

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted
IAEA-TERC-2024-01	514	100	476

Documents

S-Shape Reports:

- WWOPT: IAEA-TERC-2024-01 summary report

Worldwide Proficiency Test Exercise Webinar: Click here

IAEA-TERC-2023-01/02 Proficiency Test

Proficiency Test (PT) on determination of anthropogenic and natural radionuclides in water, soil and simulated contaminated surface samples. This year's PT was designed to monitor and demonstrate the performance and analytical capability of the participating laboratories and to identify gaps and problem areas where further development is needed. The sample set was formulated to support these goals.

- Sample 1: Spiked Water, anthropogenic and natural radionuclides, gross alpha and beta (GAB)
- Sample 2: Spiked Water, anthropogenic and natural radionuclides and GAB (environmental levels)
- Sample 3: Spiked Water, quality control sample, no data reporting and evaluation
- Sample 4: Japanese Soil, anthropogenic and natural radionuclides
- Sample 5: Simulated contaminated surface sample, gamma and beta emitting radionuclides
- Sample 6: Simulated contaminated surface sample, gamma emitting radionuclide (Cs-134)
- Sample 7: Simulated contaminated surface sample, beta emitting radionuclide (Sr-90)

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted	from Number of Countries
IAEA-TERC-2023-01	345	83	316	78
IAEA-TERC-2023-02	107	58	99	54

Documents

S-Shape Reports:

- WWOPT: IAEA-TERC-2023-01_summary_report.pdf
- ALMERA: IAEA-TERC-2023-02_summary_report.pdf

IAEA-TERC-2022-01/02 Proficiency Test

World Wide and ALMERA Proficiency Test Exercises on the determination of anthropogenic and natural radionuclides in water, soil (gamma-ray spectrum analysis exercise), and simulated contaminated surface samples.

- Detailed Sample Description:
    - Sample 01 Spiked water, 500g, Anthropogenic and natural radionuclides, gross alpha and beta (GAB)
    - Sample 02 Spiked water, 500g, Anthropogenic and natural radionuclides including H-3, Sr-90, and GAB (environmental levels)
    - Sample 03 Spiked water, 500g, Quality control (QC) with known activity concentrations of radionuclides
    - Sample 04 Spiked water, 500g, Gamma emitters, and transuranic radionuclides (TRU), GAB
    - Sample 05 Simulated contaminated surface sample, 1 pc, Gamma emitters, TRU, GAB
    - Sample 06 Blank surface sample, 1 pc, Background correction
    - Sample 07 Soil spectra files, Downloaded set of files for gamma-ray spectra analysis (no physical sample).

The instructions, the Soil gamma-ray spectrum analysis exercise and the results for the exercise are available on the TERC Proficiency Test Data Portal.

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted	from Number of Countries
IAEA-TERC-2022-01	314	79	278	70
IAEA-TERC-2022-02	105	58	98	55

Documents

S-Shape Reports:

- WWOPT: download (pdf)
- ALMERA: download (pdf)

IAEA-TEL-2021-03 WWOPT and IAEA-TEL-2021-04 ALMERA Proficiency Test

Proficiency Test on determination of anthropogenic and natural radionuclides in water, Japanese bamboo and simulated swipe samples

- Detailed Sample Description:
  - Sample 01 Spiked water, 500g, Anthropogenic gamma and alpha emitting radionuclides, gross alpha and beta (GAB)
  - Sample 02 Spiked water, 500g, Anthropogenic gamma emitting radionuclides, H-3 (for intercomparison only), Sr-90, and GAB
  - Sample 03 Spiked water, 500g, Quality control (QC) with known activity concentrations of radionuclides
  - Sample 04 Japanese bamboo, 150g, Anthropogenic gamma emitting radionuclides
  - Sample 05 Spiked water, 500g, Gamma emitters, and transuranic radionuclides (TRU), GAB
  - Sample 06 Spiked water, 50g, QC sample for TRU determination
  - Sample 07a/b Simulated swipe samples, 2pcs, Gamma emitters, TRU, GAB (identical samples)
  - Sample 08 QC swipe sample, 1pc, QC sample with gamma emitters, TRU
  - Sample 09 Blank swipe sample, 1pc, Blank sample for the swipe samples

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted	from Number of Countries
IAEA-TEL-2021-03	326	78	252	60
IAEA-TEL-2021-04	118	59	97	49

Documents

S-Shape Reports:

- WWOPT: download (pdf)
- ALMERA: download (pdf)

IAEA-TEL-2020-03 WWOPT 2020 and IAEA-TEL-2020-04 ALMERA PT 2020

Proficiency Test on determination of anthropogenic and natural radionuclides in water and fish samples and gross alpha and beta estimation on simulated aerosol filter.

- Detailed Sample Description:
  - Sample 01, Water spiked with anthropogenic and natural radionuclides, 500 g;
  - Sample 02, Water spiked with anthropogenic gamma and beta emitters, 500 g;
  - Sample 03, Water as QC sample, with known radionuclide content (isotope and activity with associated uncertainty), 500 g

- Sample 04, Dried fish sample spiked with anthropogenic and natural radionuclides, sample volume approx. 150 g
- Sample 05, 06 and 07, Printed disk samples for gross alpha and beta measurement, Sr-90(Y-90), Cm-244 and mixed respectively

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted	from Number of Countries
IAEA-TEL-2020-03	257	69	215	58
IAEA-TEL-2020-04	105	59	89	51

Documents

S-Shape Reports:

- WWOPT: download (pdf)
- ALMERA: download (pdf)

IAEA-TEL-2019-03 WWOPT 2019 and IAEA-TEL-2019-04 ALMERA PT 2019

Determination of anthropogenic and natural radionuclides in water, shrimp and simulated filters The identification of the gamma-emitting radionuclides has to be carried-out by the participants. The alpha and beta emitters to be analyzed will be specified in the cover letter.

- Detailed Sample Description:
    - Sample 01 Drinking water spiked with anthropogenic gamma-emitting isotopes and Sr-89, Sr-90 (in equilibrium with Y-90, 500 g,
    - Sample 02 Drinking water spiked with anthropogenic gamma and beta emitters, 500 g
    - Sample 03 Drinking water as QC sample, with known radionuclide content (isotope and activity with associated uncertainty), 500 g
    - Sample 04 Shrimp. 200 g
    - Sample 05, 06, 07 Simulated Air Filters

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted	from Number of Countries
IAEA-TEL-2019-03	294	74	289	74
IAEA-TEL-2019-04	106	59	105	58

Documents

- S-Shape Reports:
    - WWOPT: download (pdf)
    - ALMERA: download (pdf)

IAEA-TEL-2018-03 WWOPT 2018 and IAEA-TEL-2018-04 ALMERA PT 2018

Determination of anthropogenic and natural radionuclides in water, milk powder and calcium carbonate The identification of the gamma-emitting radionuclides has to be carried-out by the participants. The alpha and beta emitters to be analyzed will be specified in the cover letter.

- Detailed Sample Description:
    - Sample 01 Drinking water spiked with anthropogenic gamma-emitting isotopes and Sr-89, Sr-90 (in equilibrium with Y-90, 500 g,
    - Sample 02 Drinking water spiked with anthropogenic gamma and beta emitters, 500 g
    - Sample 03 Drinking water as QC sample, with known radionuclide content (isotope and activity with associated uncertainty), 500 g
    - Sample 04 Agricultural soil spiked with anthropogenic gamma emitter radionuclides, sample volume approx. 250 g

- Sample 05, 06, 07 Contaminated surfaces for surface contamination measurement

PT-ID	Count Participants Registered	from Number of Countries	Count Results Submitted	from Number of Countries
IAEA-TEL-2018-03	300	75	267	66
IAEA-TEL-2018-04	98	60	88	54

DOCUMENTS

- S-Shape Reports:
  - WWOPT: download (pdf)
  - ALMERA: download (pdf)

IAEA-TEL-2017-03 WWOPT 2017 and IAEA-TEL-2017-04 ALMERA PT 2017

Proficiency Test on the determination of anthropogenic and natural radionuclides in water, milk powder and Ca-carbonate of natural origin. The identification of the gamma-emitting radionuclides has to be carried-out by the participants. The alpha and beta emitters to be analyzed will be specified in the cover letter.

- Detailed Sample Description:
  - Sample 01 drinking water spiked with gamma-emitting isotopes and gross alpha/beta, 500 g,
  - Sample 02 drinking water spiked with anthropogenic gamma emitting isotopes, Radio-Sr and Tritium Isotopes, 500 g
  - Sample 03 QC sample, water, 500g
  - Sample 04 milk powder, Radio-SR Isotopes, Cs-134, Cs-137
  - Sample 05 calcium carbonate (CaCO3) with NORM

PT-ID	Count Participants Registered	from Number of Countries	Count Results Submitted	from Number of Countries
IAEA-TEL-2017-03	253	64	225	60
IAEA-TEL-2017-04	99	62	90	58

IAEA-TEL-2016-03 WWOPT 2016 and IAEA-TEL-2016-04 ALMERA PT 2016

Determination of anthropogenic and natural radionuclides in water, biota and soil samples The identification of the gamma-emitting radionuclides has to be carried-out by the participants. The alpha and beta emitters to be analyzed will be specified in the cover letter.

Almera only: For U-series the measurement by gamma ray spectrometry should be performed, when the Ra-226 and Rn-222 equilibrium is reached in the sample (minimum 30 days after closing the sample into the Rn-tight sample container). The activity reporting for progenies of natural decay series should be based exactly on the identified radionuclide, without branching ration correction.

- Detailed Sample Description:
  - Sample 01 drinking water spiked with gamma-emitting isotopes and gross alpha/beta, 500 g,
  - Sample 02 drinking water spiked with beta-emitting isotopes (Sr-90 and Sr-89), Am-241 and Cm-244 for gross alpha/beta, 500 g
  - Sample 03 milled clover spiked with anthropogenic radionuclides (the isotopes and activity values are published in the cover letter), 100 g
  - Sample 04 spruce needles, naturally contaminated with Sr-90 and gamma emitters, 120 g
  - Sample 05 sediment containing NORM (only ALMERA), 100 g

PT-ID	Count Participants Registered	from Number of Countries	Count Results Submitted	from Number of Countries
IAEA-TEL-2016-03	281	61	256	57
IAEA-TEL-2016-04	80	51	76	49

IAEA-TEL-2015-03 WWOPT and IAEA-TEL-2015-04 ALMERA PT

Proficiency test on the determination of anthropogenic and natural radionuclides in water, brown rice and soil samples at the suggestion of the ALMERA Reference Material and Proficiency Test Workgroup during the coordination meeting held in Vienna, 2014.

The identification of the gamma-emitting radionuclides has to be carried-out by the participants. For U-series the measurements should be performed, when the Ra-226 and Rn-222 equilibrium is reached in the sample (minimum 30 days after closing the sample into the sample container, and the application of Rn tight container is recommended). The activity reporting in case of progenies of natural decay series should be based exactly on the identified radionuclide. For example please do not apply the branching ratio correction for the activity calculation of the Tl-208 isotope.

- Detailed Sample Description:
  - 01 Spiked water, 500g, Anthropogenic gamma emitters and Sr-90
  - 02 Spiked water, 500g, Anthropogenic gamma emitters
  - 03 Spiked water, 500g, Quality control sample with known massic activity of Eu-152
  - 04 Brown rice, 200g, Gamma emitters
  - 05 Soil, 250g, Gamma emitters, Sr-90, U-238, U-234 and Pu isotopes
  - 06 Soil, 200g, U-238 and U-234

PT-ID	Count Participants Registered	from Number of Countries	Count Results submitted	from Number of Countries
IAEA-TEL-2015-03	243	57	235	57
IAEA-TEL-2015-04	82	57	73	52

For more information contact us at Proficiency-Tests.Contact-Point@iaea.org

International Atomic Energy Agency

Vienna International Centre,
PO Box 100, A-1400 Vienna,
Austria
Telephone: (+431) 2600-0
Facsimile: (+431) 2600-7
Official Email